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Questions of chemistry

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RESEARCH REPORT

Questions of chemistry

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The research reported here derives from the general field of learner-centred teaching and learning, with specific reference to undergraduate chemistry. It documents the use of student-generated questions as diagnostic of their willingness to engage in classroom interactions. It explores four ways of gathering students' written questions and their relative effectiveness. It examines students' capacity to design and present 'quality questions' during phases of their learning and the extent to which these questions are indicative of particular styles of interaction in the classroom, both with tutors and with other students. The results are drawn from data collected through written questions posted into a question box, the 'hits' recorded on a computer software system, and through one-to-one interviews with a sample of 32 students. The results provide an opportunity to discuss the quality of interactions within fairly formalized systems of teaching and learning of chemistry in a university setting and to suggest further research required in this field.

Introduction

Higher education today plays out against a very particular backdrop of social and economic conditions so that, in most parts of the world, a greater percentage of learners than ever before will progress to some form of higher education. This has brought about a noticeable shift in the business of teaching and learning in colleges and universities. Not only are new demands and challenges a matter of almost daily routine, it is also clear that part of the prevailing discourse is rapidly changing. In many respects, higher learning has adopted the register of management systems in commerce and industry so that terms such as 'quality provision', 'quality assurance' and 'excellence' have led Coffield and Williamson (1997) to call this the 'industrialisation of the language' of academia.

One consequence is the need to 'provide' to society cohorts of 'quality' students (individuals) who have developed both a substantial subject specialisms and a range of transferable or 'key' skills. These key skills include such broad capabilities as good communication skills, teamwork, problem-solving, information technology and so on, together with the 'core' skill of the graduate's ability to continue to learn new knowledge, capabilities and practices (Light and Cox 2001). This, in turn, generates a tension in the practice of teaching and learning: between the more traditional format of simply 'delivering knowledge' and the need to develop and foster independence of learning, learner-centred teaching, where students develop the ability to question, generate, reconstruct and be critical of knowledge for themselves.

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The research described here explores one aspect of that tension – the ways in which lecture and classroom interactions foster student inquisitiveness and questioning. Current research in this area indicates that levels of interaction between teachers and learners in formal instructional settings can be very low. This can mean that the key and core skills mentioned earlier have little or no opportunity for development, that curiosity and inquiry is muted, that students' capacity to ask questions – and to question received wisdom – is blunted. Our work focuses in particular on undergraduate chemistry and, in this context, we recognize that university level courses in chemistry, as in other subjects, commonly take place in very formal settings and embody expectations of teachers and learners that can constrain the variety of learning styles that are more commonly found in broader populations of learners. It seems clear to us that such constraints of formality can generate incomplete and diminished learning experiences for many students and result in under-performance in direct and non-direct learning. Situations of this kind point to the need to value learners' contributions by promoting the 'quality of interactions' (social interactions) in academic settings, which in turn, by placing the learner more centrally in the sphere of classroom activity, is expected to promote quality of learning. While talk between teacher and learner, and learner and learner, can – and often does – take place, the quality of the interaction can be very variable, either because participants are not using appropriate cognitive knowledge or because the emotional/affective dimensions of interaction too strongly influence the outcomes.

Our principal interest lies in raising the quality of teacher–student and student–student interactions in university classrooms and, in this case, university laboratories. To achieve this, our research has been designed:

i) to develop innovations in course design and planning procedures to incorporate a wide range of learning methods; and more specifically,
ii) to use computer systems to facilitate teacher–learner interactions and, in particular, encourage and explore the positive generation of questions by undergraduate chemistry students; these lead us to
iii) explore ways of providing academic support for students' questions in chemistry.

In this paper we discuss only the second and third point here: we leave the exploration of innovative course design until later. The results we describe are drawn from an initial study, are part of an on-going programme of 'work in progress', and so the outcomes we discuss are best described as 'emergent'. We present, therefore, a discussion of this key area, a rationale that is supported by some early data – it is not, at this stage a full empirical report. The central plank of this work is that the asking of general and specific questions, and the search for explanation and understanding, are two key features of high-level engagement with subject matter; in this case, in chemistry.

The nature of learners' questions

In this paper we are concerned with the questions asked by initiate scientists: university undergraduates as they embark upon a search for understanding in their study of chemistry. Our work is concerned exclusively with those questions asked by learners and not with the routine asking of questions by teachers. We follow a route

suggesting that the questions learners ask is indicative of their need for resolutions in their thinking, for understanding within the domains in which they are working and studying, and for some degree of interaction with both teachers and other students within sessions. Student-generated questions, therefore, are an important element in the teaching/learning process. First, because they can lead to improvement of understanding and retention of what a student encounters. Second, such questions can drive classroom learning and are highly effective in increasing student interest, enthusiasm and engagement. Third, learners' questions can be diagnostic of their understanding. Fourth, question-asking fosters discussion and debate

In everyday life, questions take on a multitude of forms and purposes. Ordinarily, to question is to ponder, to seek answers to a puzzle or a problem, to encounter a perplexity that requires resolution; to call something into question is to express doubts about it and to challenge its authenticity. To question somebody is to pepper them with questions, and to be questioned is to submit to someone else's insistent peppering. There are factual questions, multiple-choice questions, legal questions, rhetorical questions and, as Toulmin (1976) points out, there are philosophical questions:

> With the help of questions like these, we can look (so to say) into a mirror that shows us the workings of our own minds. (p. 4)

The approach we prefer can be summarized by reference to Dennett's (1991) notion of 'epistemic hunger'. Human beings are 'informavores' such that they have a need to 'make meaning' and understand their surroundings. The human brain, says Dennett, does everything it can to assuage epistemic hunger, to satisfy curiosity in all forms of life. However, there are two necessary ingredients. First, one needs curiosity; an openness to what is new or puzzling. Second, one needs a 'spark. Such a spark occurs when one is piqued by a detail that obstinately refuses to fit the conventional pattern, or a chance remark that somehow resonates with ones own unexplicated views or feelings. For scientists, the stimulus is often an unexplained detail or incongruity. The spark will not ignite unless it comes into contact with a body of knowledge of the right kind. That is, one cannot be curious *in vaccuo.*

In their seminal book *Teaching as a Subversive Activity*, Postman and Weingartner (1969) describe good learners as those who ask 'meaningful questions'. To some extent, encouraging students to ask meaningful questions of their teachers is still mildly subversive – even in the twenty-first century. Within the sciences, where question-asking is often accorded the status of high art, it can nevertheless be an unwelcome challenge to authority for a novice to question an expert.

Understanding is a common goal in education and recent work (Watts and Alsop 1995, Watts, Alsop, Gould and Walsh 1997, Watts, Gould and Alsop 1997) has shown that learners' questions are an essential expression of their understanding of science. As Newton (2000) points out, it is self-evident that understanding is not only 'a good thing for the process of learning, but that understanding itself is an indicator of the quality of learning'. For Sutherland (1994), understanding is the key feature that distinguishes education from training. In a particular field of study, be it chemistry or history, understanding implies the ability to recognize faulty reasoning, to construct hypotheses that go beyond the evidence available and to identify the kinds of evidence that will verify or falsify the hypotheses. Training and

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the imparting of skills can legitimately fall well short of that. Gibbs (1992) considers quality learning experiences to be:

> . . . the development of students' intellectual and imaginative powers; their understanding and judgement; their problem-solving skills; their ability to communicate; their ability to see relationships within what they have learned and to perceive their field of study in a broader perspective, to stimulate and *enquiring*, analytical and creative approach; encouraging independent judgement and critical self-awareness. (p. 20; emphasis added)

Barnett (1994), focusing on higher education, maintains that 'understanding is a central and irreducible concept'. That said, however, little is understood about how understanding comes about within individuals or groups of learners. Understanding is a very personal reorganization of knowledge: it is an economical way of knowing, that captures innumerable particulars about the world and reduces them to a coherent, manageable – and even satisfying – order. It is not a skill or competency that can be passed or transmitted from one person to another. A first supposition here is that learners' questions may be one way of assessing their level of understanding since these may show:

1. Relevant and coherent structures in learners' thinking.
2. The degree of integration of new elements and their relationship with other aspects of knowledge.
3. The learner's ability to produce an explanation, or to make a prediction, to evaluate a situation or position or even to solve a problem.

A second supposition is that learners' questions can foster and promote understanding in chemistry. Thus, it is expected that the positive generation and use of learners' questions is a vehicle for developing the quality of learning. By engineering the teaching and learning environment, and presenting learners with the possibility of asking questions easily of their teachers, it may be possible not only to gain some appreciation of their levels of understanding as they work, but also to support and facilitate learning as it happens.

A growing number of educators now emphasize the importance of students' questions in both teaching and learning for understanding, and the number of investigations looking for ways to stimulate students to generate questions is growing (Commeyras 1995, Rosenshine, Meister and Chapman 1996, Maskill and Pedrosa de Jesus 1997a, Watts et al. 1997, Marbach-Ad and Sokolove 2000). Studies at different educational levels and contexts generally indicate that learners avoid asking questions (Susskind 1969, 1979, Dillon 1988, Pedrosa de Jesus 1991). However, there is also strong evidence that if 'good' conditions are created (appropriate conditions conducive to the generation and asking of student questions) then students are willing to ask meaningful questions (Pedrosa de Jesus and Maskill 1993, Maskill and Pedrosa de Jesus 1997b). In general, learners will ask questions where they have high levels of self-confidence and self-esteem within the learning context, and where their questions are seen (Watts et al. 1997) to be valued. In some cases, asking even poorly formed and tentative questions can indicate an active, interrogative attitude that not only seeks appropriate information and opinion, but also allows some determination of the worth of what is read or heard.

Quite clearly, people can be differentially curious, differentially willing to ask questions: the 'epistemic hunger' of some being more readily satisfied than that of

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others. So, some learners ask questions easily, others less so. Some do so on the smallest contact between problem and knowledge; others require more information first before they can form a question. As discussed more fully later, the possibility exists that questioners might fall into two 'camps of curiosity':

1. Those who ask questions easily, early in the process, who take risks in the first few moments when a puzzle is formed.
2. Cautious questioners who take time, information and a lot of consideration, before they ask questions.

This may relate to people's general personality learning style and their ability to tolerate uncertainty. Some learners are able to live with uncertainty and doubt, feel relatively self-contained and can make good guesses without the need for explicit questions and answers. Others may need satisfying answers and have a need for certainty. They ask questions to minimize doubt and to restore an inner calm. Moreover, the physical and emotional environment that the learner inhabits clearly affects the nature and quality of the questions they ask. All, however, can be taught and encouraged to ask questions, and will do so increasingly if the context and 'conditions conducive for questioning' are right.

Within this, we recognize that, while a learner may frequently be able to raise questions, may be even eager to ask them, the act of uttering or presenting a question to the scrutiny of others may not be a straightforward process. A number of personal, social, psychological 'blocks' may intervene and not allow him/her to externalize these. Some researchers have taken this to mean that some learners will commonly avoid asking questions so that Graesser and Person (1994), for example, assert that:

> It is well documented that student questions are very infrequent and unsophisticated. (p. 104)

This perspective maintains that student-generated questions are normally shallow, short-answer, questions that address the content and interpretation of explicit material. Our interpretation is different: it pays particular attention to the situation in which students are interacting with teachers and with each other in order to facilitate the generation of high-level questions. These involve inferences, multi-step reasoning, the application of an idea to a new domain of knowledge, the synthesis of a new idea from multiple information sources, or the evaluation of a new claim.

While classroom contexts at university level are different from those at primary and secondary school level, there are still some similarities in, for example, the frequency and the quality of students' questioning behaviour. Understandably, the expectations of university undergraduates can be seen to be much higher and more demanding than elsewhere in the educational system. At the same time, however, they are also considered more mature and more independent as learners – so implying that they should be able to cope with the basic social interaction of asking questions. In our experience, the vast majority of first-year undergraduate students need to learn how to work autonomously and to use critical thinking skills. They need appropriate scientific stimulus as well as support in various ways, including those from the affective domain. Self-confidence, self-assurance and self-esteem are important ingredients in developing self-directed learning. Where levels of confidence increases (i.e. where students have more self-confidence and a greater

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level of interaction with the teacher), it is expected that the kind of blocks or barriers referred to earlier may be weakened and students will interact more frequently and the nature of the interactions will also be different.

The overall paucity of student questioning is not in dispute – this has been documented over time and in different settings, for example, in primary classrooms in the UK (Elstgeests 1985) and in secondary classrooms in the USA (Carlsen 1991, Commeyras 1995). The disagreement, instead, is with the implication that learners cannot routinely ask sophisticated questions of teachers – where they are enabled to do so. The critique here is of the 'atmosphere' of traditional lecture and tutorial rooms that inhibit the natural asking of questions, where learners hesitate to ask anything that might cast them in poor light with teacher and other students, where revealing of a misunderstanding is to render the student vulnerable, open to embarrassment, censure or ridicule. The emphasis in this work is that, with technique, encouragement and opportunity, learners' questions can be a fruitful means of increasing student engagement with the learning of chemistry.

Not all questions are of equal value, a notion that introduces the need to delineate clear criteria for what can be seen to be 'quality questions'. The process of question generation, and, in particular, the design and use of quality questions, is often considered an exercise in critical thinking and in the development of critical thinking skills (Fisher 1990, Browne and Keeley 1998). As we discuss later, once delineated, our interest then lies in exploring the extent to which these quality questions are indicative of particular kinds of questioners and, in this arm of the work, the relationship this might have with students' learning styles.

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Questions and interactions in chemistry

Taber and Watts (2000) chart some of the problems of undergraduate chemistry. The number of entrants to pre-university chemistry in the UK is falling, as in other parts of Europe. It is seen as a difficult and arcane subject area lacking in appeal (Watts 1999). Kee (1997) sees chemistry's declining fortunes to be a teaching problem created by the tendency towards dogmatic factual transmission rather than teaching for understanding, in this case:

> whatever passion for the subject our students have when they arrive, we, the teachers of university chemistry, have become adept at snuffing it out . . . [facts are] presented to them as they have been for the past 30 years – i.e., dry lifeless, passionless and devoid of relevance to the real world. (p. 95)

In general, we see this to indicate a 'clash of pedagogical cultures' – between those who advocate change and progress in teaching and learning towards more learner-centred approaches, and the predominant ethos of traditional 'transmission' teaching. While there are many who would explore innovative and novel ideas to engage students fully in the learning of chemistry, the widespread use of testing and assessment at all levels – leading both towards admission to colleges and universities, and as output measures once there – continues to reinforce traditional patterns of teaching and learning.

One part of the debate relates to how learner-centred principles inform the ways in which subject matter is broached. For example, teaching towards these principles pre-supposes that the teacher holds a 'dynamic' view of chemistry, seeing it as a discipline that is continually changing and undergoing revisions. It is this way that

the 'senior chemist' in the room can feel comfortable with encouraging students, as junior chemists, to make contributions, capitalizing on their remarks and incorporating students' ideas into the body of the sessions. In many respects, chemistry has a distinct advantage here. It is an interdisciplinary subject that provides an important understanding of our material world at the molecular level. It shares important ties with biochemistry, biology, pharmacy, environmental sciences along with many other disciplines, and its most central and practical objective is to synthesize new forms of matter. In this sense, it is present in and clearly has an enormous impact upon virtually all aspects of everyday life as, for example, in the production of pharmaceuticals, pesticides, fertilizers in agriculture or novel materials for the electronics industry. In this sense, chemistry is an extremely practical science, and is continuously undergoing change as new advances are made: it need not be seen simply as a static and abstract body of knowledge to be transmitted wholesale from teacher to student.

While abstract theoretical and mathematical matters are vital to chemistry, it is its very 'worldliness' that also allows it to be contextualized quite readily within real and practical contexts and applications. This 'context rich' nature of chemistry provides many opportunities to promote interaction, discussion and debate between teachers and learners, to embed the subject within the lived experience of students. One way to explore the 'lived experience' of learning in undergraduate classrooms is to examine the interactions that might take place between teachers and learners. Cunningham (1999), for instance, has made use of the 'language of industry' noted earlier to stress the role of interactions in what he calls 'strategic learning'.

In traditional models of higher education, he says, the teacher may interact with a particular learner (labelled B on figure 1) – during a lecture, asking the student to respond to some point or other – and the learner responds (A). The teacher will also relate to other learners in the same way (C). The relationship between learners (D), however, is often largely ignored or deemed unhelpful, a diversion, an unwelcome intrusion or, in some cases, a sign of poor etiquette and/or disrespect for the teacher. In an atmosphere conducive of question asking, however, the process can work

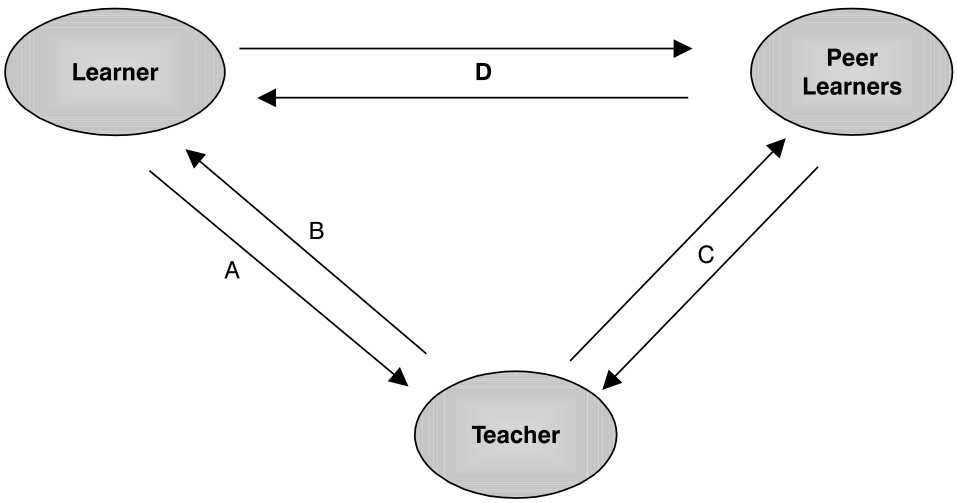


Figure 1. Channels of classroom interaction.

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differently. First, the learner's question can precede the teacher's opening remarks so that the initiation of learning comes from the learner (A). Second, the interaction between learners (D) is regarded as a key part of the learning process and it is to be encouraged. The process with other peer learners occurs in the same way.

'Worldliness' is not a quality that is exclusive to chemistry. It does, however, mean that chemistry is fully embroiled in issues of commerce, industry, social necessity and environmental impact. This social dimension of chemistry provides a dynamic arena through which the teacher can stimulate dialogue and debate – through which classroom interactions can be developed, enabling learning that contributes significantly to learners' critical social and cultural awareness. The appropriate selection of materials in lectures and tutorials can lead naturally to strong and sustained interactions between teachers and students and between students themselves. For example, it follows from a more learner-centred approach to the teaching of university chemistry that students might be engaged in some design of their own learning, might work on assigned themes in mini-projects, might form small research groups to become experts on varying topics. They might then conduct short seminars so that they share their expertise with other members of the group – a series of peer-tutorials that involve individual responsibility for constructing knowledge linked to communal sharing. Classroom discourse in these sessions includes students questioning of each other, questioning of the teacher, critiques of the issues on offer and considerable discussion as individuals contribute to and appropriate key ideas. In this forum, learning becomes an act of interpretation and negotiation with other individuals. In contrast, traditional teachers tend to default towards a much more objectivist perspective: the lecture and demonstration are the preferred modes of 'delivering' knowledge to learners. Teachers see themselves as representatives of canonical science, as practitioners who must model the intellectual skills and dispositions that students are to learn. Language is taken to be a neutral precise tool to describe the real world and effectively map knowledge as an entity that is transmitted unchanged from orthodox texts to the minds of the learners. Questioning is one way, from teacher to learner, as a means of checking that the ideas have in fact been delivered accurately and received intact, soliciting correct answers to convergent questions, providing immediate feedback on the adequacy of student response. Where questions do arise from the learner, these are commonly in a form that allows for confirmation and validation of the knowledge being transferred.

That said, there have always been problems too with 'progressive' learner-centred education; not least the impracticalities at classroom level, low levels of teacher enthusiasm and/or competence, and a general conservativism in educational circles. The key barriers for teachers in the classroom have included the difficulties of creating and adapting particular curricula to meet the needs of specific learners, managing more active classrooms and dealing with issues of accountability regarding individual student learning. A further difficulty can lie in the structure of teaching patterns in undergraduate chemistry. In many parts of the world, for example, it is not unusual for the structure of undergraduate chemistry programmes to provide for lectures to large audience groups. Audiences may be comprised of students from a range of courses related to mainstream foundation – with all the attendant problems that entails. It is possible that students' learning experiences are comprised of a mixture of teaching formats led by a range of teaching staff and in several constellations of peer and classroom groupings.

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In this vein, the research we report here relates to an initial study, conducted with a group of 32 first-year undergraduate students, drawn from the 1000 students attending an introductory chemistry course at the University of Aveiro, Portugal, during the academic year 2000–01. The structure of teaching pattern for this programme provides lectures for some 130 students at a time, an audience comprised of students from a range of courses related to mainstream foundation chemistry but who would later specialize in their final degrees. More focused teaching takes place in seminar-tutorial sessions where groups of 32 students cover issues with a specialist lecturer. Laboratory sessions are run for groups of 16, and these are often supervised by teaching assistants and technical staff. In this way the teaching is undertaken by a number of academic staff, who work hard to ensure that the programme is coherent, to diminish any fragmentation and to create good interpersonal interactions with students.

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Gathering student-generated questions

The project has been developed in full collaboration between educational researchers and the Chemistry Department, as cooperative ventures in exploring approaches to teaching and learning. Our initial focus here is on an experimental group of 32 students, with the same teacher responsible for both the duties of lecturing and for the theoretical-practical class. By developing questioning it was hoped that individuals would get to know each other better, improving the quality of the interactions, increasing the ease with which questions might be asked so that collecting and answering their questions would be enhanced. The experimental group comprises students from Environmental Sciences, Chemical Engineering, Biology, Physics and Chemistry, and Mathematics, taught by only two chemistry teachers, and therefore with a privileged relationship with the lecturers, mediated by the questions they were able to ask.

The work relies on four main components in the collection of student-generated questions:

1. a software system,
2. an e-mail correspondence system,
3. a laboratory Question Box, and
4. project workbooks.

Further data within is being collected through the outcomes of the sessions (students' tests, assignments, etc.) and also through a series of one-to-one interviews with students. The first two of the listed components highlight the use of 'new' computer technologies. Light and Cox (2001) have listed some positive as well as negative effects of the use of communications and information technology in teaching and learning. They discuss these effects in terms of the promotion of interactive learning, increases in written output, opportunities for reflection, the access to multiple frameworks/discourses/perspectives, the acquisition of computer skills and opportunities for 'learning by doing'. There are opportunities, too, to develop self-skills, to take control of one's learning, for dialogue with wider groups, and to increase collaboration between teachers and learners and between learners.

While we have some opportunity to explore some of these issues, our key interests lie in the use of software systems to facilitate question-asking. The intranet system was developed for limited access through a series of computer terminals

within the chemistry department, in the laboratories, tutorial rooms and the interconnecting corridors, thus giving relatively free access to chemistry students. The software allows those students who have use of Internet facilities outside the university to work at a distance from the department and to access the system through the use of an appropriate password. The system is fully coded so that students can follow through the various options with ease.

Our supposition has been that, with the advance of new technologies and its wider use in teaching and learning, it is possible to extend and enhance the potential for the basic types of social and human interactions through which teachers and students have traditionally engaged, and address the 'quality' of learning. For instance, interaction B (between teacher–learner) in figure 1 may take place during traditional lectures and tutorials – arenas not commonly open to easy interaction – and might now be incremented by the use of e-mail systems that can also induce interactions at A. Similarly, interaction C (between teacher and learners), usually through lectures and seminars, can by optimized by video-conferencing. Interaction D (learner–learners), traditionally enacted through projects and laboratory work, can now be encouraged by the use of Internet and intranet systems. These new technologies clearly provide different opportunities for developing interpersonal contexts. However, Mason (1998) notes that only highly motivated independent learners, good at self-pacing, computer literate and interested in computer-mediated environments are likely to make the most of them.

Our data collection began at the beginning of Semester 1, at the start of November 2000, and continued until the end of Semester 2, in the middle of June 2001. The start of the first semester was used mainly to develop, test and evaluate classroom tools, chosen with the aim of providing students with as many as possible opportunities to easily register their written questions. A short period of time was used to explain the project, to introduce them to the use of these tools and to describe the ways in which the assessment system would be used to reward question-asking. The lecturer negotiated with the students at the outset that, as an incentive, all questions generated would be judged and weighted as a positive factor on their final evaluation. The lecturer used the flexibility inherent in the grading system to award 'top-up' marks to students' other assessment grades to reward the generation of good questions. Students could gain a maximum of three points to add to their semester score (normally an average of points from several assessed elements of the course on a range of 0 to 20). Where a student had a score of 15, then he/she might gain a maximum bonus of 3 points through their asking of good questions, a score of 16 made them eligible for 2 bonus points, a score of 17 allowed for 1 bonus point – meaning that no student could exceed 18 points for the semester through this system. It is normal practice on the course to gain scores of over 18 only after submitting to an oral examination with lecturers. Both this system and the e-mail tagging of questions meant that question-asking in this pilot study could not be an anonymous affair.

In addition to their other classes, the pilot group (Class D1) met each week with their lecturer to discuss the written questions that had been generated by them in that week, providing an opportunity for answers to be presented and explored in further detail. This additional tutorial session was voluntary for the students, although all chose to attend. It was a challenge for the lecturer in terms of attempting to deal with the range of questions – and was clearly important in the development of understanding of chemistry in the group. During the second

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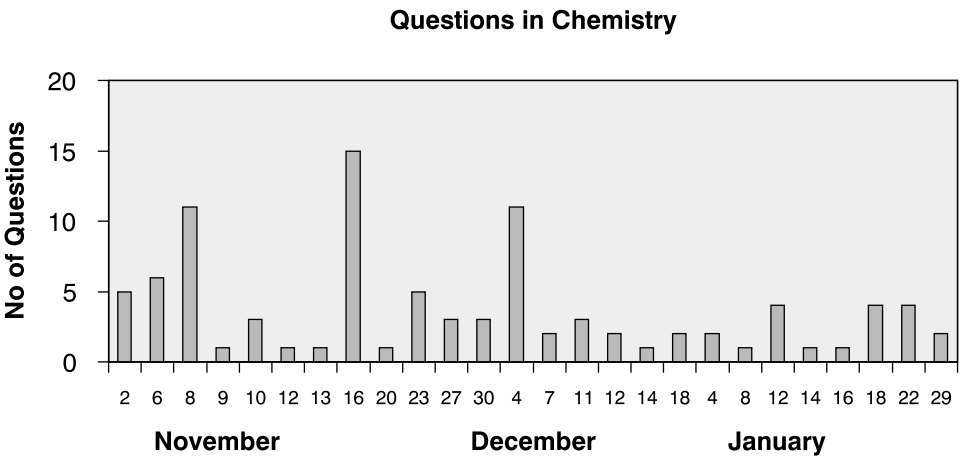


Figure 2. The number of questions asked by students each day of the module Chemistry I in the first semester.

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semester, new teaching strategies were explored, with the intention of observing the effects on students' questioning behaviour and aimed at consolidating analysis and interpretations of previous results. This allowed us both to determine the quality of the questions asked and to explore the characteristics of the questioners involved.

The graph in figure 2 shows the total number of questions asked by the pilot group over the period of the first semester within the project, which is called 'Questions in Chemistry'.

There are two issues to note here. First, the most frequently used way that students posted questions was through the classroom Question Box and, of the three methods, the laboratory workbooks were used least. Second, the general level of questions generated runs fairly evenly at about three questions per session throughout both semesters. Apart from the 'peaks' we discuss later, this overall number of written questions seems quite low – despite the very positive inducements for students to submit questions. From the interview data, we were able to discern that the asking of questions was sufficiently novel and unusual that most students were still deterred form doing so as an everyday occurrence. In Semester 1 there are three peaks, on 8 November, 16 November and 4 December. The first of these was most probably stimulated by first attempts to explore the system, and to test the extent to which question-asking worked within the computer intranet. This novelty of the emphasis on questions acted as a stimulus and prompted a first rush of activity.

The session on 16 November was a lecture for 130 students, and here it was clearly the topic and presentation that was provocative of questions. In this instance, the topic related to a theoretical discussion of molecular architecture and organization and both the difficulty and abstract nature of the issues involved generated a large number of questions. The questions on 4 December followed a particularly complex practical activity in the laboratory, and these questions arrived through the students' laboratory notebooks.

In the second semester the overall number of questions each day, as shown in figure 3, remains similar to Semester 1 bar the 'surge' on 19 April. The reason for

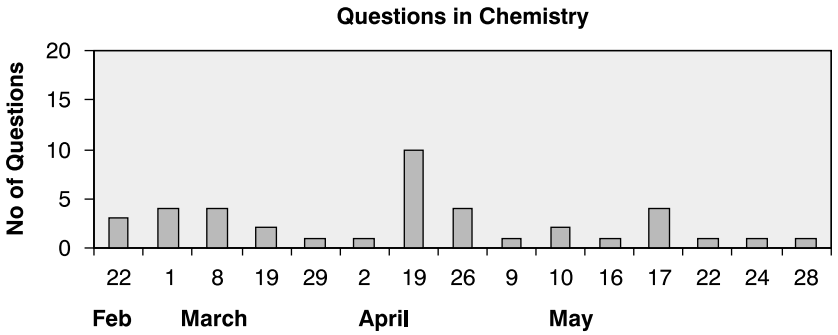


Figure 3. The number of questions asked by students during module Chemistry II in the second semester.

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this relates to the introduction into the course of a series of mini-projects that the students needed to complete by the end of the semester – students raised a number of questions surrounding the topics they were expected to investigate.

What emerges from these two graphs is that it is clearly possible to create a 'questioning environment' where asking questions (and receiving answers) become an integral part of everyday transactions between teachers and students. Throughout the two semesters the students maintained a steady flow of questions, although these could be stimulated on occasions by other events.

The key issue here is the extent to which the pilot study impacted on student performance. In this instance, we have assessment grades for the end of Semester 1 only and so results are very tentative indeed. Figure 4 shows the grade scores for the module 'Chemistry I', out of 20, for the 1000 students in the whole chemistry first-year cohort. This shows the expected normal distribution of scores across the cohort. Figure 5 shows the end-of-semester scores for the 130 students in one sub-group (Class 1), from which the pilot group of 32 was chosen. Again, this shows a broadly normal distribution of scores.

The final graph in figure 6 shows the scores for the 32 students in the pilot group (Group T1D). Here there is a small but detectable shift towards the upper

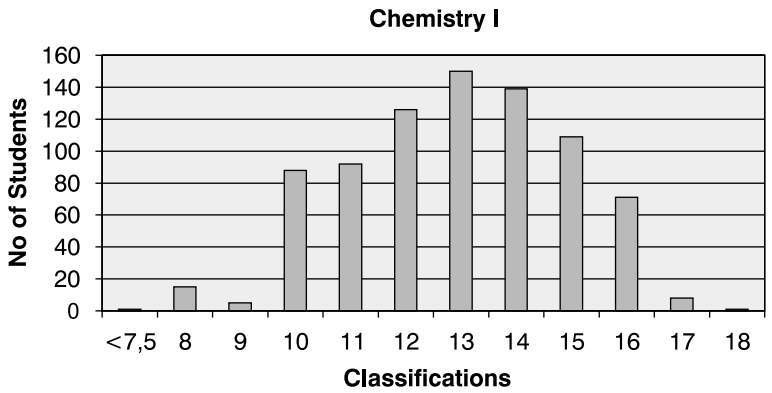


Figure 4. End of Semester 1 scores for all students for the module Chemistry I.

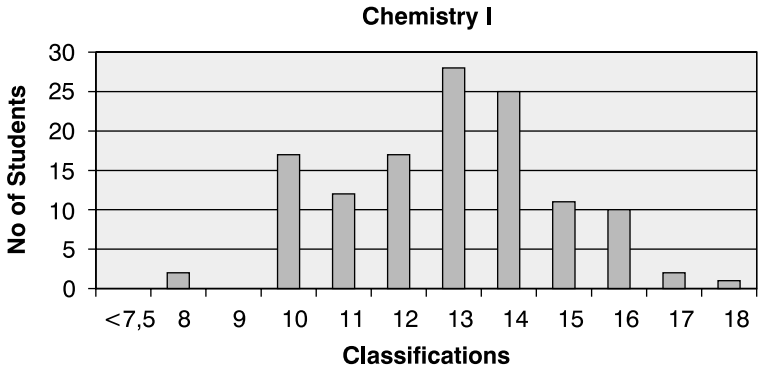


Figure 5. End of Semester 1 scores for the module Chemistry I for students of Class 1.

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grades, indicating that some of the students gained higher scores than might otherwise have been anticipated. As yet we have only supposition on which to work – there is further data to be processed that can shed light on what the reasons might be.

Whatever the reasons – whether due to the act of asking questions, the greater interaction with the teaching staff or the assessment methods used to reward question-asking (or combinations of all three) – we can be pleased that the focus of the project has not noticeably served to hinder students' progress. The pilot group has responded positively and, while there is much work yet to be done to untangle the various issues within their responses, the research undertaken so far augurs well for increasing teacher–student interactions.

Quality questions

What kind of questions could be considered 'quality questions'? What kind of properties or qualities do they have that allow them to be considered better or worse than others? Graesser and Person (1994), for example, make the distinction

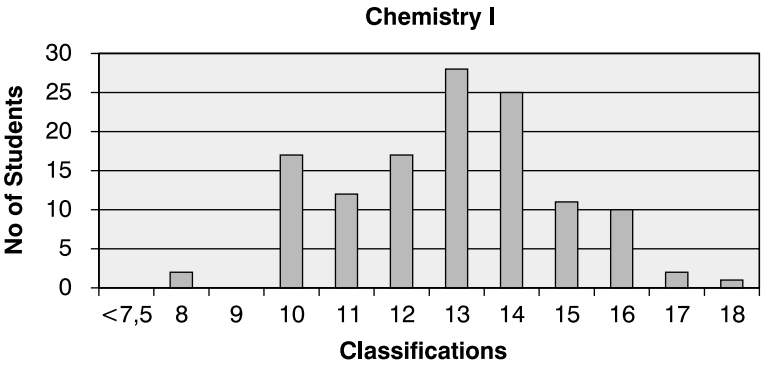


Figure 6. End of Semester 1 scores for the module Chemistry I for the pilot group (T1D).

between shallow, short-answer questions that address the content and interpretation of explicit material, and high-level questions that involve inferences, multi-step reasoning, the application of an idea to a new domain of knowledge, the synthesis of a new idea from multiple information sources, or the evaluation of a new claim. Similarly, Elder and Paul (1999) maintain that good quality questions should demonstrate clarity, accuracy, precision, relevance, depth, breadth and logic.

In the same vein, King (1990) reports positive achievements from a process called 'reciprocal questioning' where students are taught to ask each other 'high level' questions. King compared these kinds of questioning sessions with equal time open-ended discussions and, while the latter often gave longer answers, they were almost all classified as 'low level'. The questioning groups were seen to be superior in critical thinking and high-level elaboration.

A key difficulty with 'levels' such as these is that they are unipolar and value directional: asking higher level questions is clearly better (superior) than asking low-level ones. What this kind of taxonomy does not allow for, however, are notions of context, situation, task, preference, intention, strategy or goal. Our research is embedded within teaching and learning in higher education and, while arguably these notions are important in any discussion of learning, they certainly cannot be ignored in this particular forum.

More appropriate for our purposes, therefore, are bi-polar constructs to describe questions, where each pole has adaptive value so that the quality of the questions asked would depend on the nature of the situation; the learner's preferred style of working and the requirements of the task in hand. For example, Fisher's (1990) design and use of quality questions has related to 'critical thinking skills', so that he sees learning to think critically to encompass how to question, when to question and what questions to ask. He advocates teaching students 'generic' questions, such as:

> What is the main idea here?
> How would you compare this with . . .?
> But how is that different from . . .?
> Now, can you give me a different example?
> How does this affect . . .?

Browne and Keeley (1998) take this a step further by defining critical thinking in terms of the awareness of and ability to ask 'critical questions', such that critical thinking is the:

> (i) Awareness of a set of interrelated critical questions,
> (ii) Ability to ask critical questions at appropriate times, and a
> (iii) Desire to actively use these critical questions. (p. 2)

Two examples of such 'critical questions' are to ask of a task, discussion or argument 'What are the value conflicts and assumptions involved?' and 'How good is the evidence?' Following this direction has led us to develop taxonomy of questions, which distinguishes between 'Confirmation questions' and 'Transformation questions', as depicted in figure 7.

Both begin from a basis of meaning-making, of attempting to construct frameworks of understanding. *Confirmation questions* are those that seek to clarify information and detail, attempt to differentiate between fact and speculation, tackle

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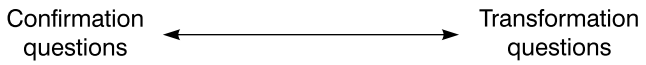


Figure 7. A spectrum of question types.

issues of specificity, and ask for exemplification and/or definition. These confirmatory questions are an attempt to decide which information is pertinent, check what basis it has for inclusion within a particular setting, and/or to determine the place and worth of particular data or evidence. Drawn from our project, some instances of such questions are:

> What is the definition of acid rain in chemistry?
> What is an example of the causes of acid rain?
> Are there (methods) ways to confirm the existence of fermions and bosons?
> What are the reasons for always pouring acid into water rather than the other way round?
> Why is it important to top up the acid in a car battery with water?
> Are they real entities or are they just labels?

Transformation questions, on the other hand, seem to signal some re-structuring or reorganization of the student's understanding. The student seems to want to get further 'inside' the ideas, to be hypothetico-deductive, to seek extensions to what is known, to cross knowledge domains. These questions explore argumentative steps, identify omissions, examine structures in thinking, and challenge accepted reasoning. Instances of these questions are:

> What might be the ways in which we could prevent acid rain?
> In your experience, what are the effects of acid rain in Portugal?
> What are all the different kinds of fields that exist?
> What is the use of this categorization?
> How is it possible for an eel to produce an average potential difference of 700 V along the 1m length of its body?
> How is it possible to generate an image from the phenomenon of nuclear magnetic resonance?
> In what ways could we increase the efficiency of the automobile engine? Using what materials, with what properties?

These two kinds of question complement each other, both are necessary and revolve around the task in hand. To pan for intellectual gold there must be something (water, sand, gravel, etc.) in the pan to actually evaluate. In some instances confirmation may precede transformation, so that transformation takes place based on the information achieved. In other cases the process may be reversed, so that confirmation is sought after a period of transformation has taken place. Both types of question are important and are inter-related.

To illustrate this point, consider the case of a person who is suddenly plunged into a situation where he/she knows very little of the topic under discussion and has to quickly make some sense of the matters at hand. There may be only a limited opportunity to ask questions before time (or patience) runs out, and the questions asked must therefore maximize meaning-making. The optimal blend will be of confirmatory questions to identify, marshal, clarify and evaluate the key issues and

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a series of searching, transformatory questions to modify the person's own arguments, transfigure understanding and to test this re-appraisal.

A set of quality questions, then, is that combination of questions that most readily enable a learner to make meaning of the learning task. Quality questions are an efficient tool kit containing the minimum most effective tools to generate good understanding. If, as some research indicates (for example, Graesser and Person 1994), student questions are commonly very infrequent and unsophisticated, then this seems an argument for encouraging students to generate a mix of questions that best suits their purpose. The next section looks a little more closely at the kind of qualities possessed by a 'quality questioner'.

Quality questioners

This section brings a shift in emphasis. Our purpose in the project has been not only to explore the social interactions that take place within undergraduate chemistry education, but also to facilitate understanding of chemistry through the generation of student questions. That is, we are setting out to foster quality questions in the way we have defined these earlier. We want to equip students with those abilities, preferences and temperaments that best contribute to learning through the asking of quality questions.

In this sense, such abilities will be uni-polar and value directional: having lots of these abilities is more highly prized than having less. From what has been said already, our delineation of a 'high-quality questioner' is that he/she has:

1. An appreciation that 'quality questions' can be of different types.
2. A feel for when questions are appropriate and inappropriate, and a sense of the roles questions can play in different contexts.
3. An ability to engineer, shape and manipulate a question.
4. The ability to appreciate the domain from which a response might be made.
5. A feel for the 'power' of questions.

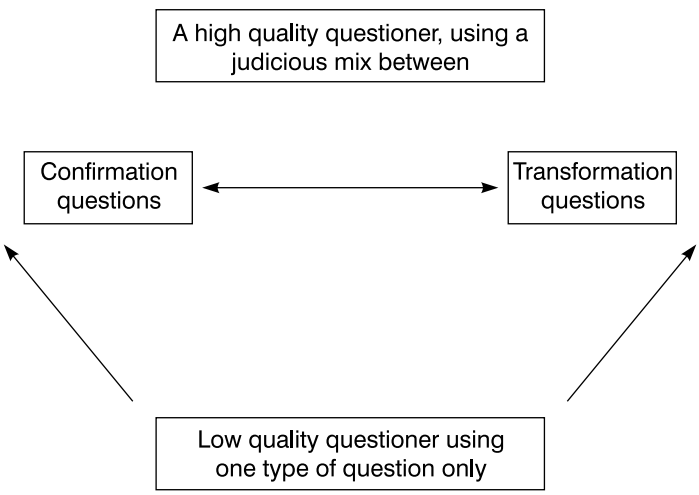


Figure 8. Quality question types.

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In contrast, a 'low-quality questioner' is a person who:

1. Asks only one type of question.
2. Asks inappropriate questions out of context.
3. Asks messy and confusing questions.
4. Lacks understanding of appropriate domains.
5. Does not recognize the 'power' of questions.

This is illustrated in figure 8.

The empirical issue that now arises is whether, from the initial data collected from within the project, it is possible to identify a 'high-quality questioning student', a 'low-quality questioning student', and perhaps one who may lie between these two levels. Our response is to portray Rosa as the first, Luis as the second and Joana as the third.

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Rosa: a high-quality questioner

Rosa asks a large number of questions in a number of forms – principally through the computer intranet system. Her questions are good ones, though they come relatively quickly. Three examples of her questions are:

> How are artificial aromas (smell, flavour) produced? For example, how can banana flavour be produced experimentally without using the fruit itself for its production?
>
> How is it chemically possible to cause rain to fall artificially, so that it can fall in any part of the world, even in the middle of a desert?
>
> Given that X-rays are so harmful to the human body, why are they used so extensively as an auxiliary diagnostic medium in medicine? Are there not better alternatives to X-rays?

Rosa seems to have intuitive insight into the area of study and can ask intelligent questions easily – she seems confident that they are meaningful and likely to move her thinking in some way, to 'prise open' the points that puzzle her. She is socially confident and does not mind being exposed in her question-asking.

Luis: a low-quality questioner

Luis seems uncomfortable when asking questions and is disinclined to ask them either verbally or in written form. They are few and slow to arrive, and then seek only basic information or descriptive responses. Two examples of his questions are as follows:

> What are properties of state? Why is energy a property of state and not heat and work?
>
> Why are atoms having a low energy electronic configuration more stable?

He is passive and solid within the group, content to take notes and undertake assignments competently, but without overt curiosity or inquisitiveness. He is composed and interested but can live quietly with any doubts or queries he has about the subject matter.

Joana: an intermediate-quality questioner

Joana is confident and spontaneous, and enjoys taking risks. She is well received in the group, alert and quick. She is willing to expose herself to criticism and

challenge. Her questions are numerous and come in several forms, so that she is easily able to explicate her uncertainties, although they lack penetration or exploration. Her questions tend to seek information, description and exemplars. Three examples are:

> What is the difference between transmittance and absorbance?
>
> How can we represent graphically the maxim absorbance values of the two solutions (blue and red)?
>
> What is the difference the hypothetical and molecular structures?

Discussion

It is important to note at the outset that we have been dealing here with written questions submitted through three 'collection points', and through which the identity of the question-asker is known. Our preliminary results (Pedrosa de Jesus, Teixeira-Dias and Watts 2001) show that these undergraduate students have been very positive about this new kind of interaction with their teachers. During interviews and informal conversations they have found the emphasis on generating questions to be an interesting and challenging departure from the norm. The data we discuss in this paper show that they took good advantage of this innovative strategy, with the majority asking meaningful questions – very few indeed resorting to playful or mischievous questions. The teachers involved in the course were also keen to see what emerged from the pilot study and worked hard to ensure that students had opportunities to post their questions. All the questions were directed at the teacher – none of the students took the opportunity to e-mail or use the Question Box to ask questions to each other. This inter-student element of the study will be explored through other means.

What emerges from this work is that it is clearly possible to create a 'questioning environment' where asking questions (and receiving answers) becomes an integral part of everyday transactions between teachers and students. Throughout the two semesters the students maintained a steady flow of questions, although these could be stimulated on occasions by other events. Some learners do ask questions easily, others less so. Some seem to do so on the smallest contact between problem and knowledge, others require more information first before they can form a question. Of course, 'high-quality', 'low-quality' and 'intermediate-quality' questioners may lie in either camp, as we illustrated with our cameos of Rosa, Luis and Joana.

The number of written questions generated runs fairly evenly at about three questions from the group per session throughout both semesters, with a few points at which there is a significant rise in the number posted. The start of the study saw a large number of questions arriving, with the novelty of the emphasis on questions acted as a stimulus that prompted a first rush of activity. At other points, it was clearly the topic and presentation that was provocative of questions. In one instance, the topic related to a theoretical discussion of molecular architecture and organization, and both the difficulty and abstract nature of the issues involved generated a large number of questions. A second instance followed a particularly complex practical activity in the laboratory and these questions arrived through the students' laboratory notebooks. A third instance related to the introduction into the course of a series of mini-projects that the students needed to complete by the end

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of the semester – students raised a number of questions surrounding the topics they were expected to investigate.

This research shows that it is possible to change the 'atmosphere' of traditional lecture and tutorial sessions in the teaching of university chemistry, the atmosphere that inhibits the natural asking of questions. We have shown that, with technique, encouragement and opportunity, learners' questions can be a fruitful means of increasing student engagement with the learning of chemistry. For whatever reason – whether due to the act of asking questions, the greater interaction with the teaching staff or the assessment methods used to reward question-asking (or combinations of all three) – we can be pleased that the focus of the project has not noticeably served to hinder students' progress, and to create a mood of understanding and motivation. The pilot group has responded positively and, while there is much work yet to be done to untangle the various issues within their responses, the research undertaken so far augurs well for increasing the quality of student-generated questions.

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